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DEPARTMENT FOR PROMOTION OF INDUSTRY AND INTERNAL TRADE (DPIIT)

APPLICATION FOR COPYRIGHT REGISTRATION — FORM XIV

DOCUMENT 04

SOFTWARE TECHNICAL DOCUMENTATION

Comprehensive Engineering Architecture, Component Hierarchy, Routing, State Management, and Telephony Integration Specification

Statutory Category:	Computer Software / Literary Work (Section 2(o), Copyright Act, 1957)
Software Title:	ARB SoftTech AI Enterprise Voice Calling Agent Platform — Frontend Software
Application Package:	voxora-enterprise-ai-voice (Version 2.0.0 Production)
Languages & Tech:	TypeScript (v5.6.3), JavaScript (ES2022), React 18.3.1, Vite 6, Tailwind CSS
Target Authority:	Registrar of Copyrights, Copyright Office, New Delhi, India
Statutory Mandate:	Deposit under Rule 70(5), Copyright Rules, 2013 (First 10 & Last 10 Pages Format)
Date of Compilation:	October 2026

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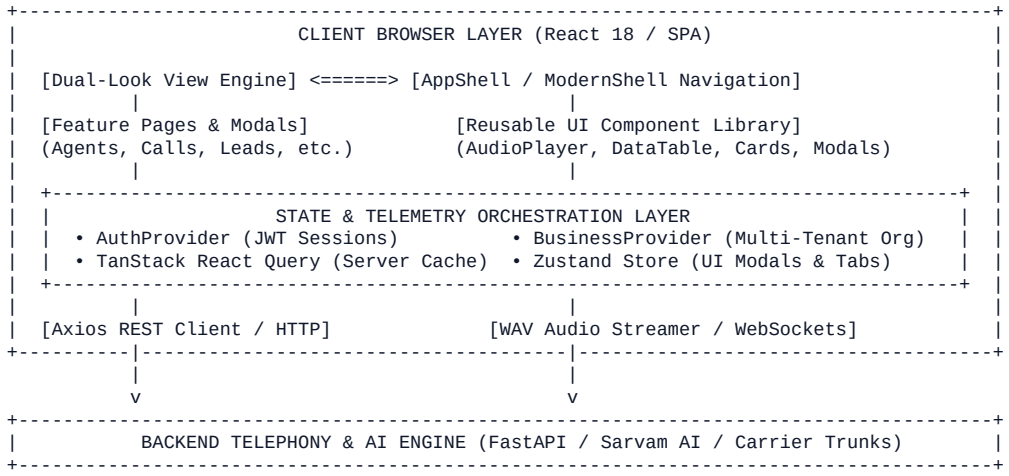
1. HIGH-LEVEL SOFTWARE ARCHITECTURAL DESIGN

The frontend application of the **ARB SoftTech AI Enterprise Voice Calling Agent Platform** is architected as a high-throughput, reactive Single Page Application (SPA) powered by React 18 and Vite 6. It acts as the presentation, orchestration, and telemetry layer for the underlying telecommunications backend (FastAPI, Sarvam AI telephony engine, and VoBiz/Twilio PSTN carriers).



2. FRONTEND DIRECTORY STRUCTURE & CODEBASE ORGANIZATION

Directory Path	Module Role	Key Files / Submodules
src/app/	Application Root & Routing	App.tsx, router.tsx, providers/ (Auth, Business, Theme, Query, Toast)
src/components/layout/	Layout Shells & Navigation	AppShell.tsx, Sidebar.tsx, Header.tsx, LookToggle.tsx, QuickDialModal.tsx
src/components/ui/	Atomic UI Library (21 components)	AudioPlayer, Badge, Button, Card, DataTable, Drawer, Modal, Tabs, Select
src/components/effects/	Modern Visual Effects	AmbientBackground.tsx, GlowProvider.tsx, ParticleStars.tsx, ambient.css
src/features/agents/	AI Voice Agent Studio	AgentsPage.tsx, AgentFormDrawer.tsx, AgentWorkspacePage.tsx
src/features/calls/	Telephony Call Monitoring	CallsPage.tsx, ActiveCallsPage.tsx, CallDetailDrawer.tsx, QueuePage.tsx
src/features/leads/	AI Lead Intelligence Pipeline	LeadsPage.tsx (Gemini intent scores, customer details, why-reasons)
src/features/questionnaires/	Conversational Questionnaire Flow	QuestionnairesPage.tsx, QuestionnaireEditorPage.tsx, Excel/Docx Import Modals
src/features/modern/	Modern High-Tech Design Suite	ModernDashboard.tsx, ModernCalls.tsx, ModernLeads.tsx, ui.tsx (17 views)
src/features/settings/	Telephony & AI Credentials	SarvamSettings.tsx, VoBizTelephonySettings.tsx, TwilioTelephonySettings.tsx
src/services/api/	REST & WebSocket Client Services	client.ts (Axios), calls.ts, agents.ts, leads.ts, telephony.ts (17 services)
src/types/	Enterprise TypeScript Schemas	api.ts (Exhaustive domain models, API schemas, and response interfaces)
src/utils/	Export Engines & Formatters	agentPdfExport.ts, businessPdfExport.ts, excelExport.ts, quickDial.ts

3. DECLARATIVE ROUTING STRUCTURE & DUAL LOOK ARCHITECTURE

The application utilizes react-router-dom (v6.28.1) via createBrowserRouter. A hallmark architectural feature is the **Dual Look Engine**: routes are encapsulated within a LookPage component that dynamically renders either the classic enterprise view or the futuristic modern view depending on operator selection.

URL Route Path	Classic Component	Modern Component	Functional Scope
/	DashboardPage	ModernDashboard	Operational Telephony Cockpit & KPIs
/calls	CallsPage	ModernCalls	Historical Call Records & Audio Player
/active-calls	ActiveCallsPage	ModernActiveCalls	Real-Time Ongoing Call Waveforms
/agents	AgentsPage	ModernAgents	Conversational Voice Agent Studio
/agents/:id	AgentWorkspacePage	AgentWorkspacePage	Prompt Engineering & Tuning Workbench
/leads	LeadsPage	ModernLeads	AI Lead Intelligence & Qualification
/questionnaires	QuestionnairesPage	ModernQuestionnaires	Conversational Flow Editor & Importers
/businesses	BusinessesPage	ModernBusinesses	Multi-Tenant Enterprise Organizations
/followups	FollowupsPage	ModernFollowups	Scheduled Customer Callbacks & Tasks
/reports	ReportsPage	ModernReports	Business Intelligence & Analytics
/settings	SettingsPage	ModernSettings	Sarvam, VoBiz, Twilio, Gemini Config

4. STATE MANAGEMENT & PROVIDER CASCADE

State management in the application is architected across three distinct tiers:
1. **Server State (TanStack React Query v5):** Handles asynchronous API fetching, automated polling of active calls, intelligent stale-time caching, and optimistic UI mutations.
2. **Domain & Session Context (React Context Hierarchy):** Defined in src/app/providers/:
• ThemeProvider: Manages dark/light appearance tokens.
• QueryProvider: Encapsulates the master React Query client.
• AuthProvider: Stores operator authentication tokens, role permissions, and login modal states.
• BusinessProvider: Manages multi-tenant organization switching, persisting active business ID in local storage.
• ToastProvider: Global notification queue for feedback alerts.
3. **Ephemeral UI State (Zustand v5):** Controls lightweight, rapid-toggle interface states such as drawer expansions and tab switches.

5. TELEPHONY INTEGRATION & AUDIO PLAYBACK SUBSYSTEM

The frontend interfaces directly with telecommunications hardware and carrier trunks via backend proxy endpoints:
• **Real-Time Audio Player (AudioPlayer.tsx):** Built to stream lossless 8,000Hz 16-bit PCM WAV audio recorded directly from live telephony channels. Supports play/pause, seek scrubbers, volume controls, and turn synchronization.
• **Live Telephony Polling (useActiveCalls.ts):** Automatically polls /api/calls/active every 2,000 milliseconds, calculating dynamic call elapsed time and displaying animated waveform pulses.
• **Turn-by-Turn Transcript Synchronization:** Diarizes conversation dialogue into structured chat bubbles distinguishing the AI Voice Agent and the Customer, accompanied by speech recognition confidence and latency markers.

6. BUILD, OPTIMIZATION & DEPLOYMENT INFRASTRUCTURE

The project is engineered for production-ready, zero-downtime enterprise deployment. The compilation and delivery pipeline consists of three core components:

• **TypeScript Build Verification (npm run build):** Runs tsc -b (strict TypeScript compiler verification) followed by vite build. Rollup bundles and tree-shakes all modular source files into optimized distribution chunks in dist/assets/.

• **Multi-Stage Docker Containerization (Dockerfile):**
• *Stage 1 (Builder):* Node.js 20 Alpine environment installs dependencies and compiles static bundles.
• *Stage 2 (Production Runner):* High-performance Nginx Alpine image receives only the compiled dist/ artifacts, reducing final container footprint to under 25 megabytes.

• **Production Nginx Reverse Proxy (nginx.conf):** Directs static asset requests with long-term cache headers while proxying /api/* and telephony WebSocket streams to the backend engine on port 8001. Implements Single Page Application fallback routing (try_files $uri $uri/ /index.html).

7. SECURITY CONTROLS & EXCLUSION OF SENSITIVE CREDENTIALS

In accordance with software security best practices and statutory filing standards:
1. **Zero Hardcoded Secrets:** All carrier credentials (VoBiz Auth IDs, Twilio Account SIDs, Sarvam AI API keys, and Google Gemini API keys) are configured dynamically through UI input forms and transmitted securely to the backend via HTTP POST/PUT requests.
2. **Exclusion from Copyright Deposit:** Environment configuration files (.env, .env.local), private authentication tokens, and server passwords have been strictly excluded from this documentation package to prevent any compromise of operational credentials.

Technical Architect / Engineer:

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Senior Software Documentation Engineer
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Verified for Copyright Submission:

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Authorized Officer / Signatory
For: ARB SoftTech